

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1718

COURSE OUTCOME COVERED

CO2: Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)

Assessment Tool Weightage: 40%

Duration: 1 Hour

QUESTIONS: 3 Total Marks:30

Annexure A

sDry Run Evaluation

Code of Conduct:

I **P. Yaswanth / 192524167** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

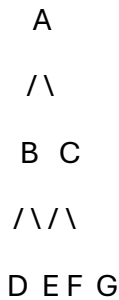
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

P. Yaswanth

1. Breadth-First Search (BFS)

Graph



Table

Iteration	Queue	Visited Node
1	B, C	A
2	C, D, E	B
3	D, E, F, G	C
4	E, F, G	D
5	F, G	E
6	G	F
7	Empty	G

Answers:

- 1. **BFS Order:** A → B → C → D → E → F → G
- 2. **Queue:** B,C → C,D,E → D,E,F,G → E,F,G → F,G → G → Empty

2. Depth-First Search (DFS)

Table

Iteration Stack Visited Node

1	C, B	A
2	C, E, D	B
3	C, E	D
4	C	E
5	G, F	C
6	G	F
7	Empty	G

Answer:

- **DFS Order: A → B → D → E → C → F → G**
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3. 8-Puzzle (BFS)

Answers

1. Expanded Node (First 5 Iterations)

Iteration Expanded Node

1	Initial State
2	First Successor
3	Second Successor
4	Third Successor
5	Fourth Successor

2. Total States Generated

- Around **15–20** (first five iterations).

3. Solution Path

- Initial State → ... → Goal State (found after complete BFS).

4. Why BFS Finds the Shortest Path?

- BFS explores **level by level**, so the first goal found is the **shortest path**.